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Internet of Things(IoT) Enabled Intelligent Traffic Control Management System By Hybrid Swarm Intelligence(SI) Algorithm

Voore Subrahmanyam,
Dept. of CSE,
Anurag University,
Hyderabad, Telangana State, India
subrahmanyam.cse@anurag.edu.in
ORCID: 0000-0002-9860-6355

O. Sri Nagesh,
Dept. of CSE,
Anurag University,
Hyderabad, Telangana State, India
nagesh.osri@gmail.com
ORCID: 0000-0003-3344-9413

Y.Venkatramana Reddy,
Dept. of CSE,
Anurag University,
Hyderabad, Telangana State, India
vrcuddapah@gmail.com
ORCID: 0000-0001-9317-7259

A.Prashanth Rao
Dept. of CSE,
Anurag University,
Hyderabad, Telangana State, India
prasanthraoit@cvsr.ac.in
ORCID: 0000-0002-5119-3987

Mandala Suresh
Dept. of CSE&AI,
SR University,
Warangal,, Telangana State, India.
mandala.suresh83@gmail.com
ORCID:0000-0003-2902-0577

G. Neelima
Dept. of CSE&AI,
SR University,
Warangal,, Telangana State, India
gneelima83@gmail.com
ORCID: 0009-0006-5716-0701

***Abstract* - The congestion of traffic is one of the problem for road transport in the city. This traffic congestion problem affects by population growth and also usage of vehicles that cause inconvenience for people facing problems like air pollution, routing, traffic congestion. The proposed system Internet of Things(IoT) based Intelligent Transport System using Hybridized algorithm with combination of Ant Colony Optimization(ACO) and Particle Swarm Optimization(PSO) algorithm more effective way to solve the traffic control by dynamically adapt as per the volume of the vehicle traffic at circles. ACO chooses best routing path in rush traffic hours for providing an optimal path in a city traffic for short distance. Particle Swarm Optimization(PSO) algorithm is an optimization algorithm to optimize the vehicle velocity, complex traffic management and congestion avoiding solver to avoid accidents, blocking of traffic and save travel time for the intelligent traffic management system in the city. The computation results shows proposed Intelligent Traffic Control Management System using Hybridized Ant Colony and Particle Swarm Optimization(ITCMSHACPSO) algorithm is provided the best optimized features to enhance performance of Quality of Service(QoS), reliability, maximize vehicle count to reach their destination, avoid congestion in a traffic, best path selection for the route, manage velocity compare with other individual algorithms like ACO, PSO.**

Keywords – Transport Management System, Traffic control, Optimization, Internet of Things(IoT), Ant Colony Optimization(ACO), Particle Swarm Optimization(PSO).

I. Introduction

At ongoing transport management system causing traffic congestion problems by more number of vehicles are using by vast growing of population. Due to the accidents and traffic congestions happening every day. It is essential to develop a method to optimize the vehicle better route as well as to control the traffic in the City premises. The various directions of the nature of traffic flow bi-directional depends on quantitative and qualitative traffic data. And also the destination of the vehicles not the same. The destination of the vehicles from origin to destination also different. The quantitative traffic data includes parameters of velocity, length of queue, waiting time etc. The qualitative traffic data includes for detecting the congestion at various junctions in the city premises. The traffic jam problems is one of the complex problem as ever growing day-by-day in the city where specially in junctions. It is not showing any improvement to solving this problem. It cause time waiting problem and pollution problem. It is an essential solve these traffic jam problem. Various technologies and methods have been taken to consider and to solve traffic congestion problem in the city. All popular technologies like IoT based sensors CC cameras, Ultra sonic sensor, GPS (global position system), RFID(Radio Frequency Identification) etc. The GPS receives is about 10 meters, RFID limited the range of covering the signals. Various solutions are proposed are CCTV cameras, GPS, Radio Frequency Identification(RFID) to control the traffic congestion problem in the city [1,2,3,4]. These technologies are involving high maintenance cost as per their technological limitations. Now it is an essential to improve with advanced technologies and with optimization algorithms for improve the precision traffic control, traffic congestion problems in the city premises [5].

II. Existing System

In this Existing system various proposed their research in a particular problem solving. Authors [6] proposed a method for control traffic in city that cause traffic congestion, reduce fuel congestion by Ant Colony Optimization.

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In this proposed paper the authors implement a method of repelling of pheromone that generate repulsive force of all vehicles for that travelling paths are distributed through a road networks. This algorithms results congestion free in a traffic.

Authors [7] proposed an intelligent traffic management system based on Random early detection using evolutionary algorithms to reduce vehicle waiting time, average trip time, fuel consumption in the city traffic. Authors used SUMO simulation tool and iREDVD technique using EA optimization tool for efficient management of traffic in the city premises. Authors shows reduces the vehicles' waiting time, average trip time, fuel consumption but not shown satisfactory optimize route.

Authors [8] proposed an efficient to minimize delay and overhead in a traffic control in city. The detection of malicious vehicle use unmanned air vehicles(UAV). A routing protocol VRU is proposed to route the data to deliver packets of data between the vehicles with UAV protocol named as VRU_vu. The packets of data the routing in between the UAVs using protocol shown as VRU_vu. Authors used NS-2.35 simulator of Ubuntu to know the performance of VRU routing in urban traffic. Mobile Similator used to know the motions of vehicles to produce UAV's movements is able to routing control of the vehicles. Authors not providing security energy saving for UAV because it is an essential to UAV's life time.

Authors [9] proposed IoT base an efficient traffic prediction and control system propose&apply OWENN algorithm integrated with Intel 80286 microprocessor to control traffic signal system for traffic management system in urban. In this proposed system apply data collection procedure by IoT, classification, IoT based traffic optimization methods. The traffic data is collected by IoT.

The traffic should optimize as per traffic direction. IBSO system optimizes IoT values while the traffic has been controlled with Intel 80286 micro processor. This proposed system obtained results shows an efficient management of traffic control system in the City. Authors shows traffic prediction and traffic signal control system for smart city but not shows the best route to control the traffic in the city.

Authors [10] proposed IoT enabled noise control& monitoring traffic vehicles was being taken authors for the proposed paper about traffic affects on noise conditions on the road. But authors concentrate only on noise emissions in the city, but not how the noise being affected by congestion of vehicles, speed of the vehicles, routing methods.

Authors proposed Internet of Things(IoT) based system design and develop traffic noise problems analyze & safety system of a vehicle in a traffic of the city. The system includes analyze sensor collected data for noise emissions of vehicle real-time effective management as well as control of traffic system in city. It is a real-time traffic control system to analyze the noise levels and monitoring of noise levels in a particular area of the particular city.

III. Methodology

Our proposed methodology of layered architecture for transport system in the city. This layered architecture from the bottom Transport layer, Network layer, Application layers and their modules show in Fig. 1. The Transport Infrastructure Layer having components like Vehicle to Vehicle communication, IoT based pollution monitor sensors like MQ135, Automation of traffic lights, CCTV cameras, Vehicle parking sensors, Ultra sonic sensors etc. In Network communication layer having network related components like Routing for vehicle travelling, IoT communication devices, 5G for mobile communication, Gateway/ Routers, Wireless sensor networks, RFID (Radio Frequency Identification)etc. The Application is topmost layers for user defined layer for purpose of the user utilization of resources. In Application layers components related to Cloud computing, Augmented reality, Virtual reality, User interface, Remote monitoring, Server/ Clients systems etc. To solve problem in transport system we introduce Hybrid ACPSO algorithm for effectively solve the routing and congestion.

The Fig. 1 is describes about the Layered Architecture for Transport system in the City for better management of traffic control the City premises. In this proposed paper the layered architecture is introduced and every layer is inter-related with each for better communication with other layers.

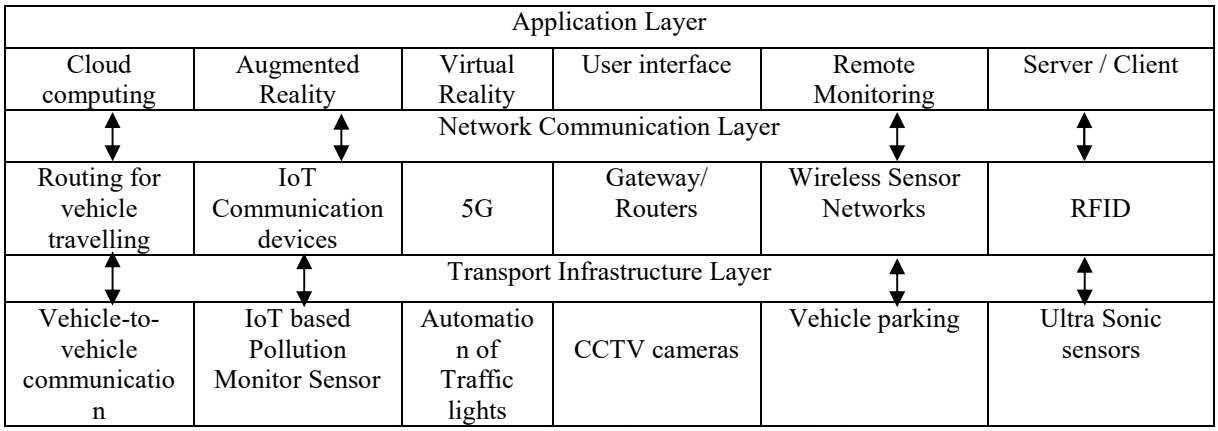

Fig. 1. Layered architecture for Transport system in City

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In our proposed research we are planning our layered architectural framework for IoT based intelligent traffic management system in the city. In this layer framework, the first layer is physical layer consists of IoT based vehicles. Second layer is network layer consist of sensors, actuators, communication systems, routing, servers. Third layer is application layer consist of cloud, AI, user interface etc.

IV. Proposed Work

The traffic control system in the city face challenges like routing, congestion. This paper solve these problems effectively by the Hybrid optimization algorithm. It is based on Ant Colony Optimization(ACO) showing best path for vehicle in a traffic congestion. The vehicles are may initially choose a various different routes, while being this situation the vehicles receive a traffic congestion, automatically choose an alternative route for smooth and congestion free route.

A. Ant Colony Optimization(ACO)

The two algorithms ACO and PSO are showing best route and solve congestion in a traffic. The ACO and PSO are main components for Hybrid optimization algorithm. The ACO is a probabilistic model i.e. inspired by studying as per conduct of ants. The behavior of real ants discovery of food and come back to the nest. While discovery of food, ants make shortest distance for route. Every ant deposit concentration of its pheromone laid on a particular route.

The other ants also following the same route as per volume of this pheromone status to find destination. ACO algorithm is one of the natural inspired algorithms we are taking as a model for solving our real-time routing problem in traffic. The congestion is shows by pheromone collected on a particular route of the road. The pheromone value is identified as per movement of vehicle in a particle path. If the vehicle enters in a particular route, the threshold value is incremented. When the road is busy, means the pheromone value in a particular route is decrements. In order to avoid road congestion the pheromone value is changed. When the conditions are normal automatically pheromone value is setback to reasonable value [11,12].

B. Particle Swarm Optimization (PSO)

Particle Swarm Optimization(PSO) is a population based meta-heuristic optimization algorithm. PSO inspired by birds. The parameters of PSO is being update position as well as velocity by applying local best & global values [13]. The position as well as velocity of PSO followed the following equations.

$$vi(t + 1) = \omega vi\ (t)+c1(pbest(t) — xi(t))\ c2\ (gbest(t)-xi(t+1)) \quad (1)$$

$$(t + 1) = t + 1 \quad (2)$$

In above Eq. (1), Eq. (2), the PSO parameters pbest is local best for a particle, gbest = the global best for a particle. These parameters are influence of the Particle Swarm Optimizatio(PSO) algorithm for personnel best (pbest) and global best (gbest) to get optimality. In PSO the pbest is the personnel best and gbest is global best for the position of any particle (i.e. bird) in a swarm (group of birds).

xi(t+1) = at iteration t of the particle's position i
vi(t+1) = at iteration t+1 of the velocity particle i
The equation1 & equation2,
t=no. of iterations,
vi(t)= particle i speed
xi(t)= particle i position

In equation1, the acceleration coefficients C1 and C2 are obtained where 0 less than C1, C2 less than or equal to 2. ∈ (0.8; 1.2) is considered for inertia weight. Inertia weight ω is 1 for update speed equation of PSO. The Inertia weight denote for balancing the local and global search of PSO algorithm. For consideration of big inertia weight much better for the search globally. For lesser inertia weight consider local search easier. Inertial play an important role for balancing between local and global search. In PSO the optimum results will achieved by less iterations.

V. Proposed ACO-PSO Hybrid Algorithm

This is a Hybrid algorithm by ACO and PSO to improve the pheromone value to reduce the traffic congestion. Every route has been assigned a initial pheromone value. This value has been changed as per the vehicles movement in this route. The PSO having parameters P_{best}, G_{best} noted as personnel best, global best solutions. ACO-PSO Algorithm having local search, global search characteristics to show best optimal route with velocity to travel the vehicles in selection of best route. The ACO algorithm, the pheromone update by ants to achieve best route. PSO algorithm uses for velocity, improve the P_{best}, G_{best} parameters to update pheromone values. To improve ACO-PSO Algorithm both ACO and PSO for best route and velocity, P_{best}, G_{best} values of vehicles for a smart traffic [14,15].

The implementation steps for hybrid algorithm providing a solution for the construction, local search G_{best} and P_{best} to update pheromone updating. If there are n ants starting from the departure. Each ant to find to construct a route by applying rules of ACO algorithm. When all n ants complete the construct the solution, in that n ants the top r best ant will perform local search for improving the solutions for fulfill the task. After that these ants are update their memory. The hybrid algorithm update the pheromone trails as per the G_{best} solution and also P_{best} solutions. If the G_{best} not improve the w consecutive iterations than hybrid algorithm performs to modify current pheromone trails. The hybrid algorithm reset the P_{best} of the ants to prevent them laying pheromone in the same route.

In this proposed paper for optimization of traffic control considering swarm intelligence algorithms. These swarm intelligence algorithms are having inbuilt features of collective behavior, adapt with nature to solving real-time problems around us. These swarm intelligence algorithms are a model of solving computational problems. These are providing effective solutions and solving optimization problems. To solving traffic control we consider Ant Colony Optimization(ACO) and Particle Swarm Optimization(PSO) algorithms. The ACO for find the shortest distance from source to destination of any vehicle to minimize route distance. The PSO is a model for traffic control and congestion. By applying hybrid model of ACO include with PSO for solving effectively of transport optimization problem.

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The proposed algorithm steps as follows.
Step1: Initialization of parameters
Step2: m ants are perform the construct the solution
Step3: Local search perform by r best ants
Step4: Now update the G_{best} and P_{best} solutions select r elite ants
Step5: If G_{best} not maximize within ω iterations than go to step6, otherwise go to perform step7.
Step6: Randomly distribute the pheromone matrix then reset the P_{best} solution than go to perform step8
Step7:Update the pheromone matrix as per P_{best} solutions of all elite ants
Step8: Whether maximum no. of iterations reached then go to perform step9 or otherwise goto perform step3 to perform next iteration
Step9:Get output for G_{best} as optimized solution

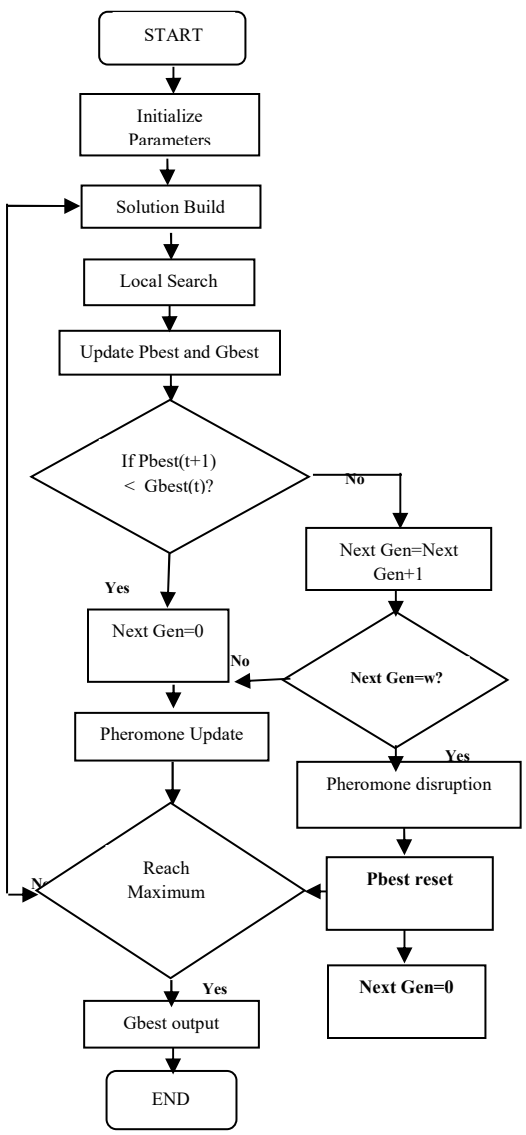

Fig . 2. Flow chart for the proposed Hybrid algorithm

In Fig. 2. Flow chart shows the proposed hybrid algorithm by combination of ant colony optimization(ACO) and particle swarm optimization(PSO) to form hybrid algorithm for better optimization of traffic management system in City.

A. Initialization Pheromone

To calculate the speed the vehicles, length as following Equation.

$$V_x = V * Cos\ \Theta$$
$$V_y = V * Sin\ \Theta \quad (3)$$

$$a_x = dv_x / dt$$
$$a_y = dv_y / dt \quad (4)$$

In Eq.(3) Vx and Vy showing velocity of vehicles simultaneously for both directions.

In Eq. (4) a_x, a_b showing latitude and longitude acceleration of vehicles. The sensor are assigned to the vehicles provide data the Equ(3), Equ(4) are taken to consideration.

The time t is used for formulation for pheromone initialization by following Equ(5).

$$\rho_{ij}(t) = \rho_{ij}(t-1) + C_1 * length(edge_{ij}) \quad (5)$$

In Eq. (5) ρ_{ij} is the value of the pheromone from vehicle i to j which is denotes as value of edge. The time is denotes as t=0 measured as time is taken every vehicle travel from source to destination. C1 is the constant value of vehicle [0,1]. The length will be increased as the number of edges increases. This is shortest and congestion free route.

B. Pheromone update process

The ants update the value of pheromone on the edges for time taken by vehicles to travel. It functioning by following Eq. (6).

$$\rho_{ij}(t) = \rho_{ij}(t-1) + C_2 * (1 + \Delta\rho_{ij}) \quad (6)$$

In above Eq. (6) $+\Delta\rho_{ij}$ = 1/N, or otherwise =0. N represents the number of nodes, C2 represents the constant value which lies between [0,1].

The time taken for travel and waiting at intersection are waiting at green light are calculate sum of travel time addition to waiting time as traffic lights in a signal area.

$$\tau_{ij} = (1 - \rho)\ \tau_{ij} + \Delta\rho_{ij} \quad (7)$$

The above equation the states about the pheromone concentration deposited in the route by ant of ant colony optimization(ACO). The pheromone is a chemical substance released by Ant while moving from source to destination in a particular route. If the pheromone quantity is more than automatically the other ants also moving in the same route.

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In above Eq. (7) τ_{ij} = amount of pheromone between path i and path j. The ρ is the amount of pheromone evaporation rate, whereas $\Delta\tau_{ij}$ values denotes the ant deposited volume of pheromone on ground. $\Delta\tau_{ij}$ is considered $\Delta\tau ij^k = 1/L_k$. The ant K travel on paths i and j. Lk is denotes cost of Kth tour for a particular length. Otherwise $\Delta\tau ij^k = 0$.

$$\Delta\tau_{ij}\ (t + n) = \rho\tau_{ij}\ (t) + \Delta\tau_{ij} \tag{8}$$

In above Eq. (8) $\Delta\tau_{ij}$ = the sum of improve the pheromone status at the edges and dissipation. The pheromone numbers per circle as shown by (t+n). The following fitness function is apply for smart traffic management system in the city. The goal of an idea of implementation is to increase the vehicles to reach their destination and minimize the time to reach their destination.

$$Fitness(F)=((\sum(a=0\ )\text{^}(A+C)\ \ W_V\ ) + C*S_t)\ /(A\text{^}2+ Cr) \tag{9}$$

The main goal of smart traffic management system is to increase number of vehicles to reach destination(A) and Arrival time(St). The certain situation the number vehicles decreasing for not arrived their destination and continue(C). The minimize overall duration of vehicle's journey(Ov). To calculate the time of vehicle stop and waiting(Wv). It is to be calculate total time required for all vehicles to be reached their destination. The number of vehicles not yet complete their journey towards to reach their destination(Cr), time spent their journey even not reach destination(St), the vehicles arrived at their destination(A^2). The traffic signals lights colour indicates three lights i.e. green(G), red(R), yellow(Y) in phase states are ph and tl are denoted as G_{lm} and R_{lm}.

$$C_r = ((\sum(l=0)\ \text{^}\ tl(\sum(m=0)\ \text{^}ph\ S_{lm})R_{lm})\ \ )\ /G_{lm} \tag{10}$$

VI. Results Analysis And Discussion

To evaluate for the performance of optimization of traffic lights for smart traffic system in the city we use following parameters taken for consideration. Dataset – We extract the real-time information from a digital map and design optimization for traffic direction of the roads and streets. Our proposed experimental approach carried by nature inspire computing algorithms ACO, PSO Hybrid ACPSO Hybrid Algorithm for optimization of traffic in the city. In our proposed model, as per number of vehicles on the road, and as per number of traffic signals operated, the number of vehicles reached to their destination has been calculated by using fitness function. As per our proposed result analysis, when comparing with ACO 22%, PSO 28% , Hybrid ACPSO 40% maximize flow of IoT vehicles achieved to reach their destination through these traffic junctions. ITCMSHACPSO(Intelligent Traffic Control Management System Hybrid Ant Colony Particle Swam Optimization algorithm). Swarm Intelligence algorithms are showing model to effectively solve our computational problems. ACO algorithm to solve routing problem in the traffic. The PSO algorithm avoid congestions, accidents to control the traffic in the City. The Hybrid algorithm combination of better features of Ant Colony optimization and Particle Swarm optimization.

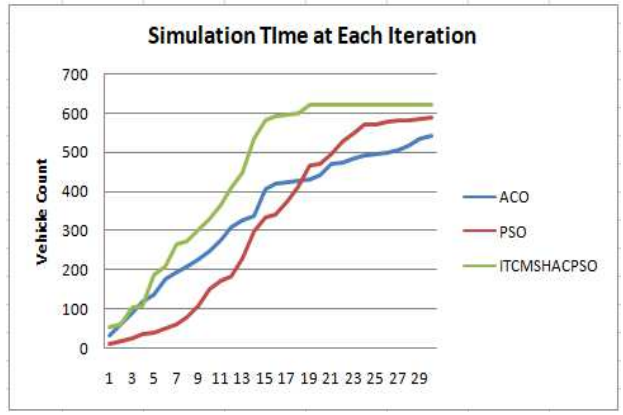

Fig. 3. Flow of vehicle using ACO, PSO, Hybrid ACPSO algorithms. X axis shows no. of iterations & Y axis shows Vehicle count.

In Fig. 3 as per simulation results it is observed that vehicles count is maximize by applying ITCMSHACPSO (Intelligent Traffic Control Management System Hybrid Ant Colony Particle Swam Optimization) algorithm, unlike individual ACO, PSO approach. The Hybrid ACPSO algorithm approach is to maximize vehicles without congestion.

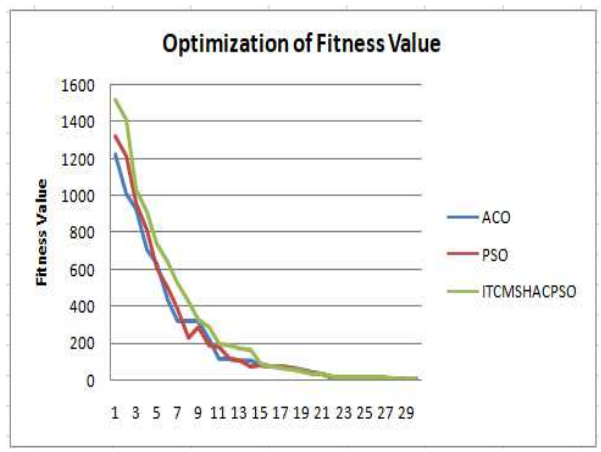

Fig. 4. Optimization of Fitness Value using ACO, PSO and Hybrid ACPSO algorithms. X axis shows no. of iterations & Y axis shows Fitness value.

In Fig. 4 as per simulation results it is observed that vehicles count is maximize by applying Hybrid ACPSO algorithm when compare with individual ACO, PSO algorithms approach. The Hybrid ACPSO algorithm approach is achieve optimization of fitness value.

TABLE I. SIMULATION PARAMTERS

Parameters	Value
Software for simulation	MATLAB
Simulation Area	20m x 20 m
Number of nodes	30
The velocity of every node	Constant
Transmission rage	Less than 100 m
Speed of vehicle	40 m/s

In the Table1 is the parameters considering for proposed model. It represented by two columns in the table i.e. parameters taken and values. The parameters values shows software for simulation, similar area, number of nodes, velocity of every node, Transmission range and speed of vehicle with showing respected parameter values in second column.

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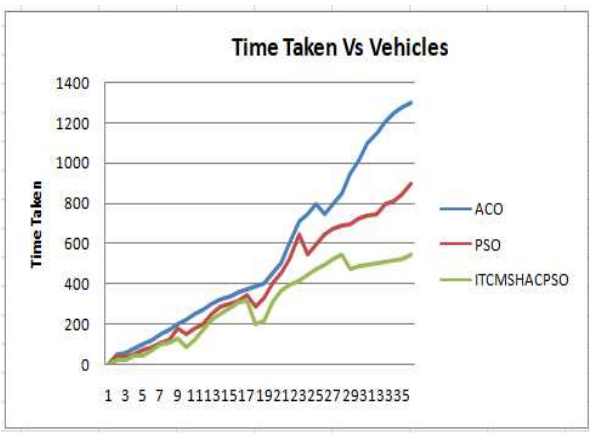

Fig. 5. Optimization of Time Taken and Number of Vehicles. X axis shows no. of iterations & Y axis shows Time taken.

In Fig. 5 as per simulation results it is observed that vehicles time considered by speed of vehicles to reach their destination. The optimization of Time is minimized to reach destination by vehicles to minimize congestion by applying Hybrid ACPSO algorithm, unlike individual ACO, PSO algorithms approach.

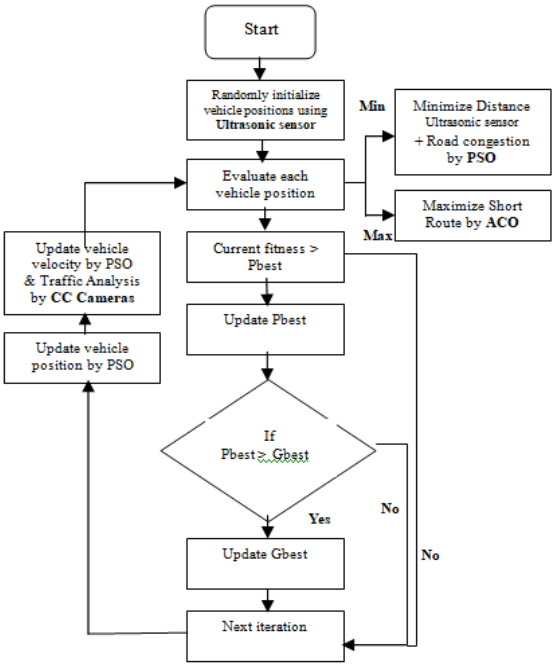

Fig. 6. Shows a complete information flow cycle of computation, to measure distance between vehicles(ultra sonic sensor), traffic analysis(CC cameras) etc. are resources of this hybrid ACO and PSO algorithm for efficient traffic management system in the city.

VII. Conclusion

The Intelligent Transport System using Hybridized algorithm with combination of ACO and PSO algorithm more effective way to manage the traffic control in city. Swarm Intelligence algorithms are one of the bio inspired model having features of self discipline, co-ordination, collective behavior to solve various computational problems. ACO chooses best routing path in rush traffic hours for providing short distance. PSO algorithm optimize the vehicle velocity, congestion avoiding to solve avoid accidents, blocking of traffic and save travel time. In this proposed paper introduce hybrid algorithm is the combination of ACO and PSO features to effectively solve the traffic control in the City.

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